

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2023 - 24 Examination

Semester: 7

Subject Code: I03105407

Subject Name: Information and Network Security

Date: 10/04/2024

Time: 02:00 pm to 04:30 pm

Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1	Objective Type Questions - (Fill in the blanks, one word answer, MCQ-not more than Five in case of MCQ) (All are compulsory) (Each of one mark)	(15)	CO	PO	BT
	1. In the AES-128 algorithm the round key is _____ bit and the Round Input is _____ bits.				
	2. The result of $7 \bmod 9 =$ ____ and $11 \bmod 9 =$ ____? a. 7, 2 b. 2, 2 c. 2, 7 d. 7, 7				
	3. Rail Fence is a _____ Technique for encryption/Decryption.				
	4. Vignere Cipher is a _____ cipher.				
	5. In Rail Fence cipher with depth=3, how many rows are used to encrypt the plaintext? a. 1 b. 2 c. 3 d. 4				
	6. Change one bit of the input ,at least 50% of the output should be different ,this property is called _____. a. Confusion b. Diffusion c. Non-Repudiation d. Data Integrity				
	7. _____ is the assurance that someone cannot deny something a. Access Control b. Integrity b. Non-Repudiation d. Authentication				
	8. Deffie-Hellmen is a _____ Algorithm.				
	9. In RSA, Plaintext M can be calculated by $M =$ _____.				
	10. The method of hiding the message is called.				
	11. Message Digest length in SHA-512 is _____ bits.				
	12. Which of the correct combination key size, no. of round for AES. a. 128, 12 b. 192, 10 c. 226, 14 d. 256, 14				
	13. Is OTP a extended version of Vernam Cipher (True / False)				
	14. What is Bruteforce?				
	15. Full form of HMAC is _____.				
Q.2	Answer the following questions. (Attempt any three)	(15)			
	A) Compare Substitution and Transposition techniques. List out different Substitution and Transposition techniques.				
	B) Explain HMAC with neat diagram.				
	C) Why AES is better than DES ? Discuss in each perceptive.				
	D) Explain one time pad (OTP). What is the drawback of OTP?.				
Q.3	A) Discuss Advanced Encryption Algorithm (AES) in detail.	(07)			
	B) Use Playfair cipher substitution technique and find out the cipher text for the following given key and plaintext. Key = MONARCHY , Plaintext= INSTRUMENTS	(08)			
	OR				
	B) How can Kerberos be used in remote authentication? What is the role of the ticket-granting server and authentication server in Kerberos?	(08)			

Q.4	A) Why are block cipher modes important, and explain cipher block chaining mode with diagrams?	(07)			
	OR				
	A) Explain meet-in the-middle attack in 2 DES. Explain Triple DES with three keys.	(07)			
	B) Provide a detailed explanation of the RSA algorithm, and illustrate its operation with a example.	(08)			